SCP-S

MEMBERSHIP

Stars in catalog	36
with Gaia GSP-Phot	19 (53%)
with radial velocity	1
Stream id (sID)	2

GEOMETRY · great-circle fit

Centroid RA, Dec	7.5, -34.7 deg
Centroid l, b	346.6, -80.8 deg
Pole RA, Dec	93.5, 4.8 deg
Length (φ1, 2–98%)	28.6 deg
Width σ(φ2), robust	0.32 deg

KINEMATICS

pmRA* med ± σ	1.94 ± 0.45 mas/yr
pmDE med ± σ	2.50 ± 0.60 mas/yr
μφ1* med ± σ	-2.28 ± 0.56 mas/yr
μφ2 med ± σ	2.22 ± 0.53 mas/yr
Parallax (cat.) med	0.092 ± 0.245 mas
Parallax (wtd, ZP-corr.)	0.103 ± 0.022 mas
Distance	d = 9.7 ± 2.1 kpc
Physical length	4.83 kpc
RV median ± σ(MAD)	47.3 ± 0.0 km/s
RV gradient dV/dφ1	—

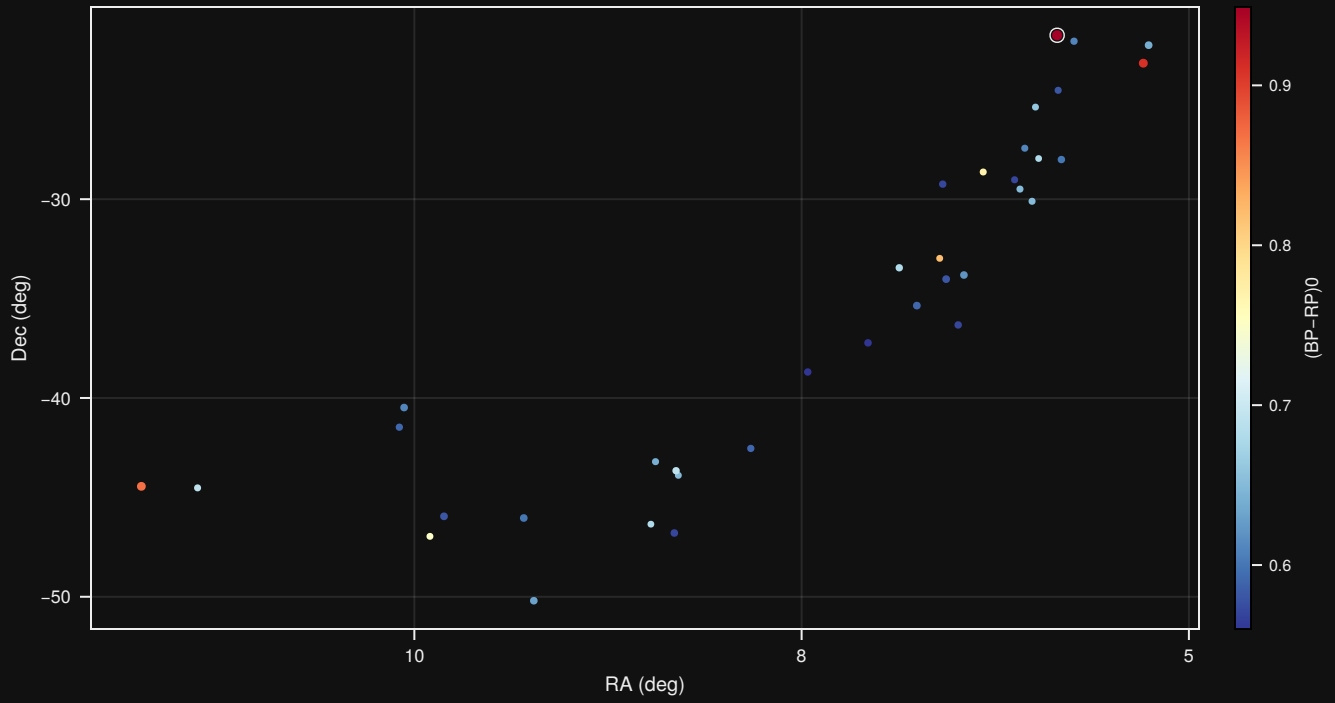
PHOTOMETRY · POPULATIONS

G0 range (median)	14.7–19.9 (18.9)
(BP−RP)0 median	0.62
GSP-Phot [M/H] (n=19)	-1.59 ± 0.77
GSP-Phot Teff median	6356 K
RUWE > 1.4	0.0 %
PM clusters (largest)	2 (20 stars)

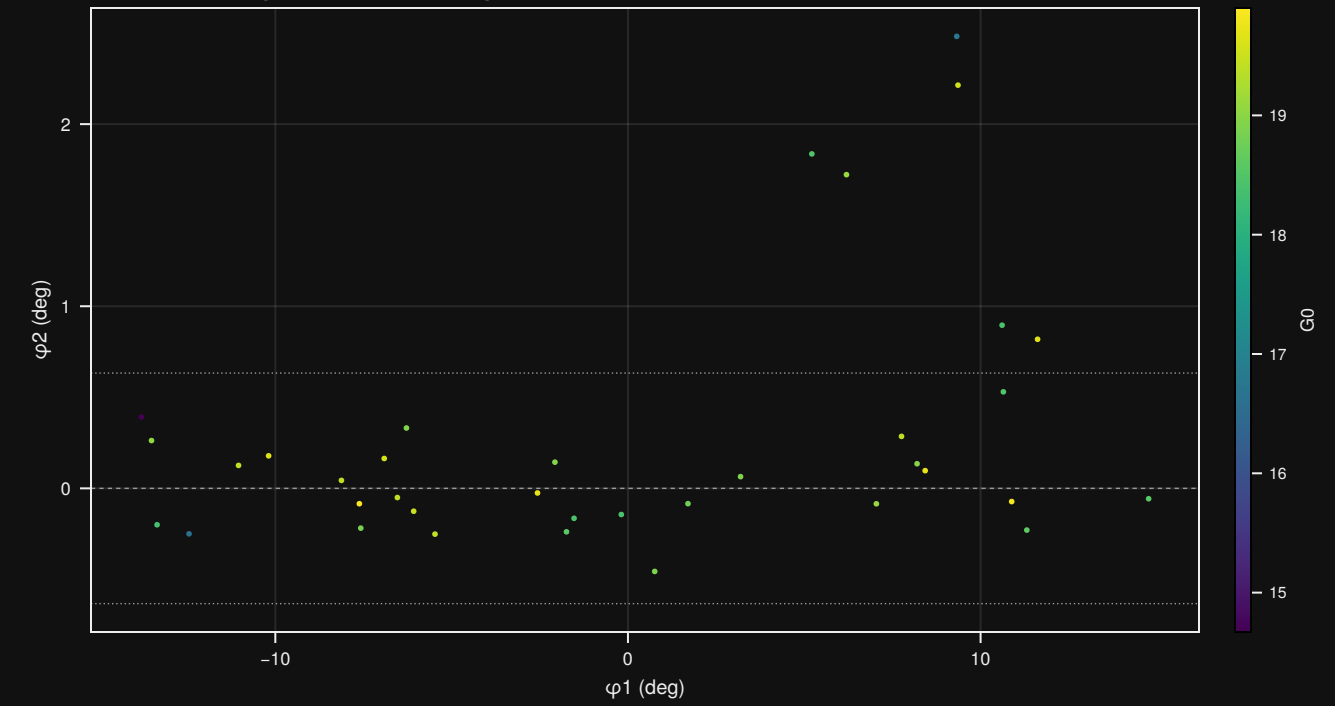
IBATA+2024 · TABLE 3

n catalog | paper	36 | 36
n_RV catalog | paper	1 | 1
<[Fe/H]> paper	-2.22
Stellar mass (paper)	530 Msun
Pole offset vs paper	0.9 deg

Sky map · catalog members (n = 36) · size ∝ brightness · ring = has RV



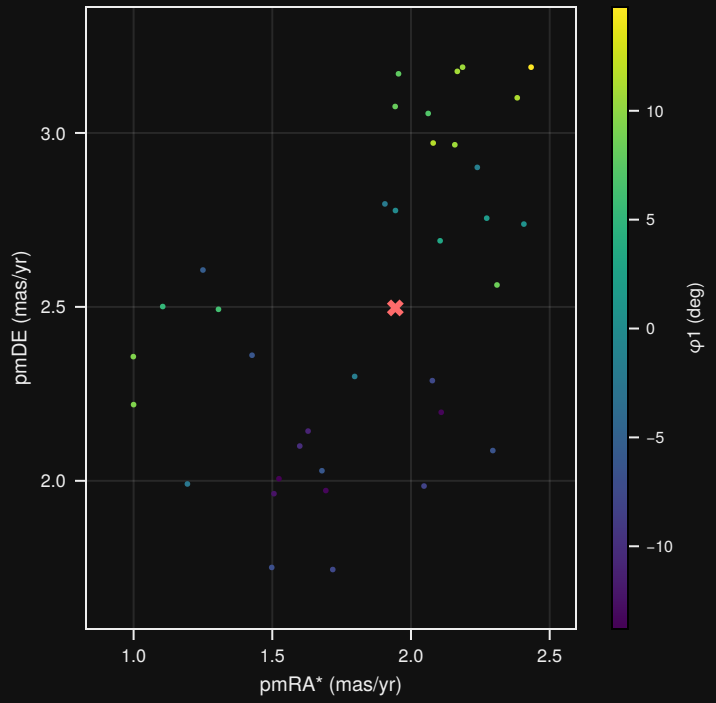
Stream frame (great-circle fit) · length 28.6° · width σ 0.32°



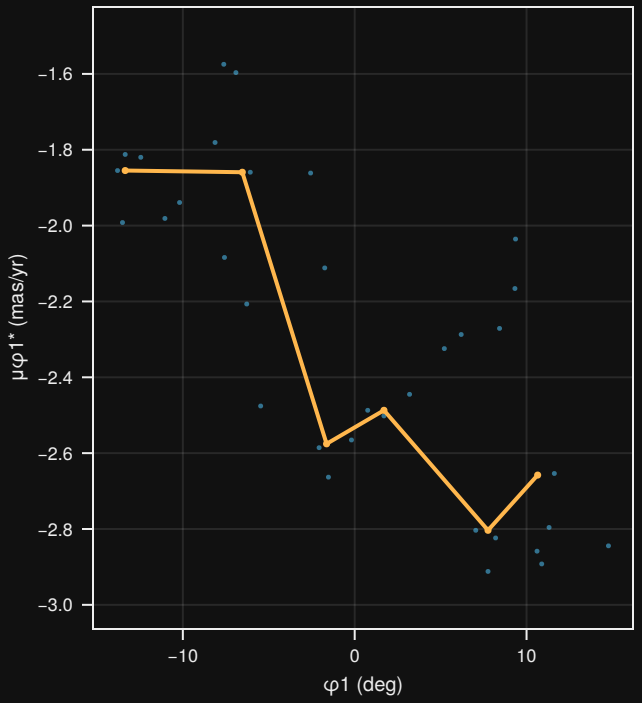
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Kinematics · proper motion · parallax · radial velocity

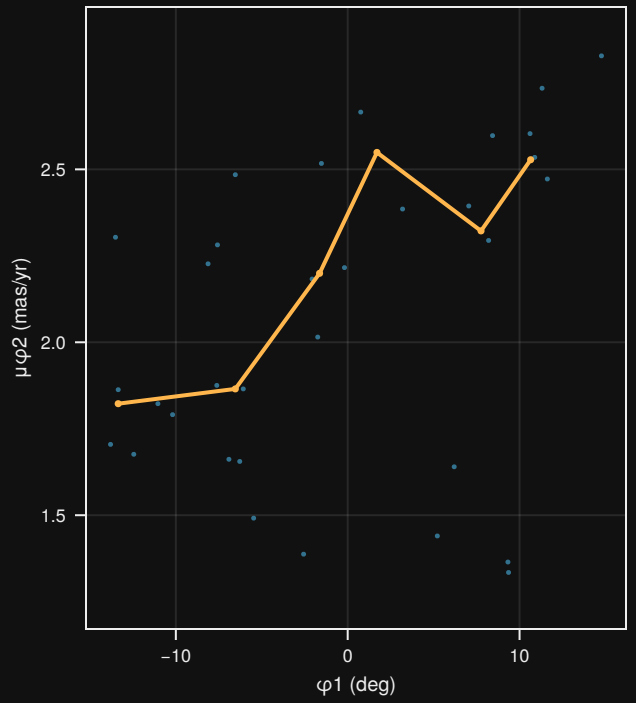
Proper motion (n = 36)



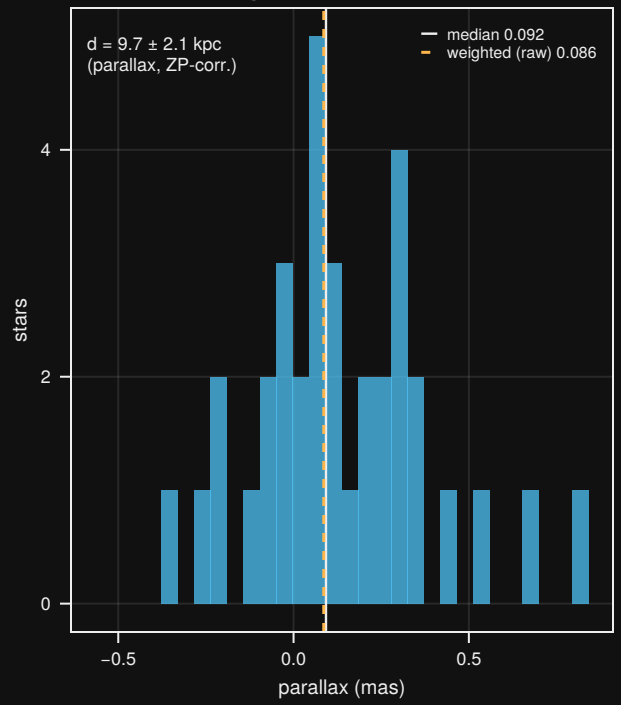
μφ1* vs φ1 · σ = 0.56



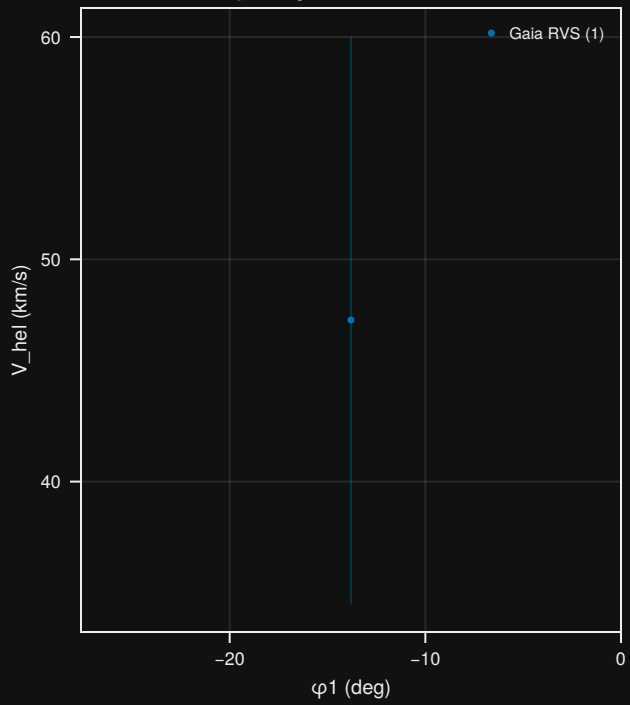
μφ2 vs φ1 · σ = 0.53



Parallax (catalog, n = 36)



Radial velocity vs φ1 (n = 1)



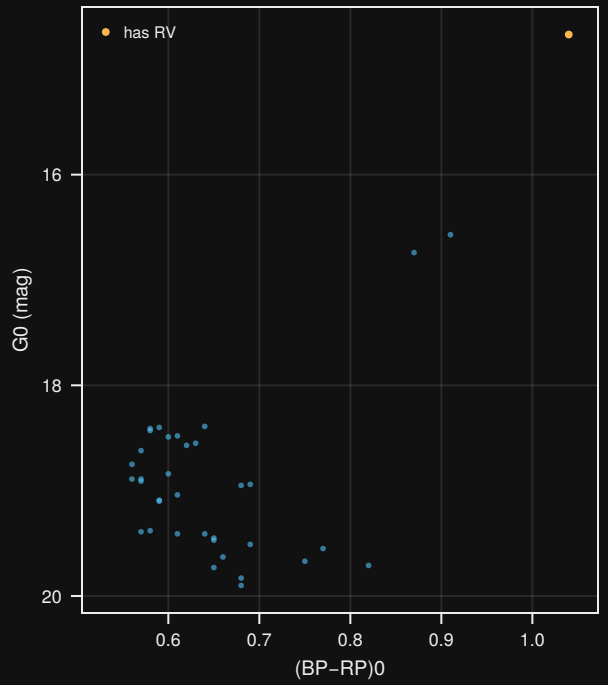
V_hel distribution

n_RV = 1 (< 3)

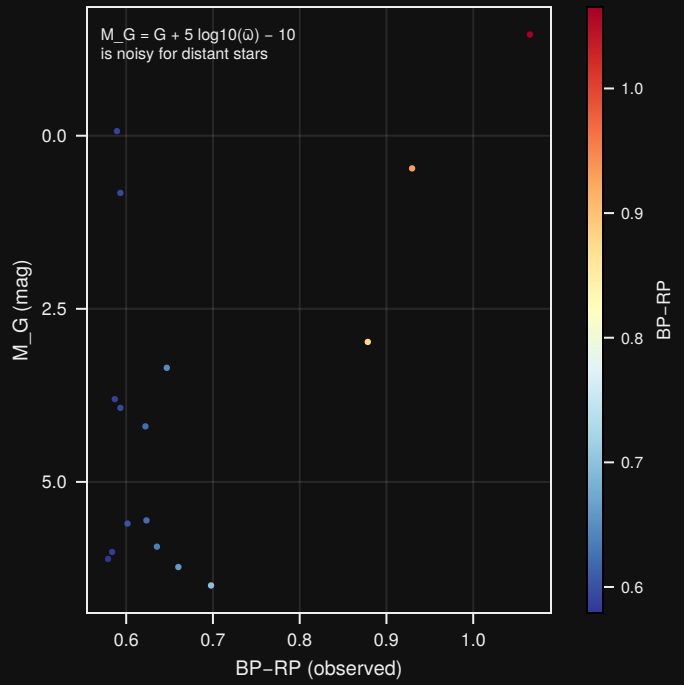
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Photometry · stellar types

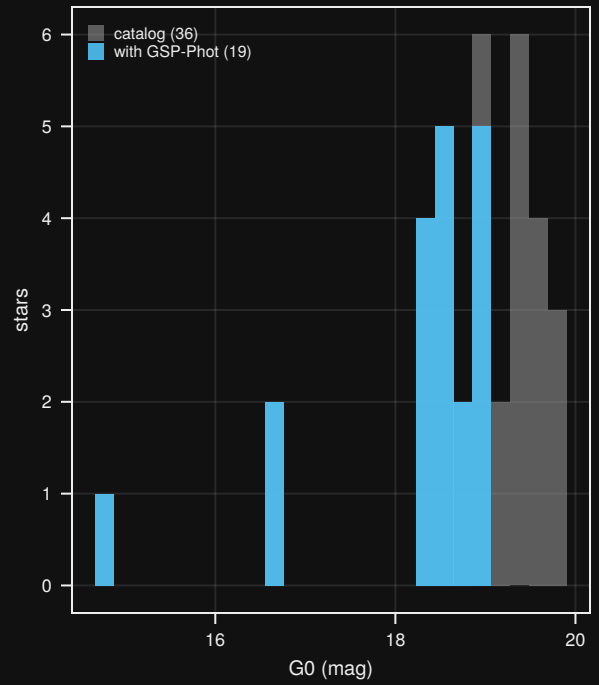
CMD · dereddened (n = 36)



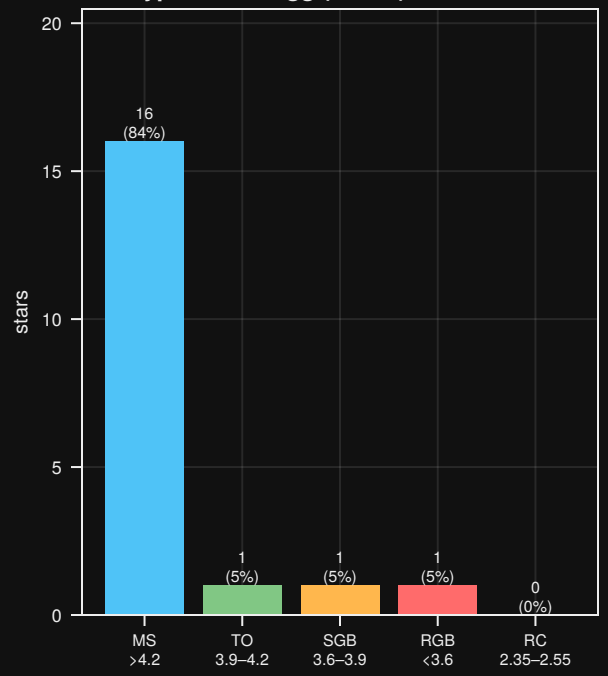
HR diagram · ϖ>0 (n = 16/19)



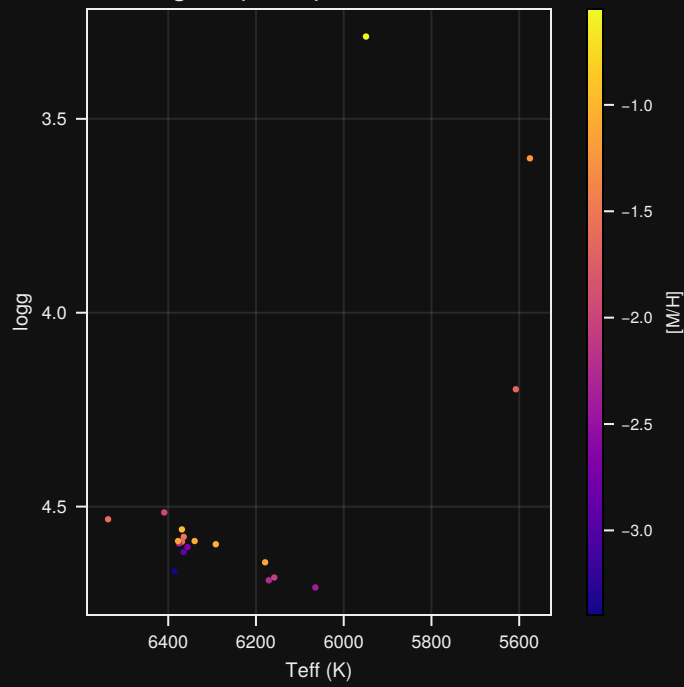
G0 & GSP-Phot coverage



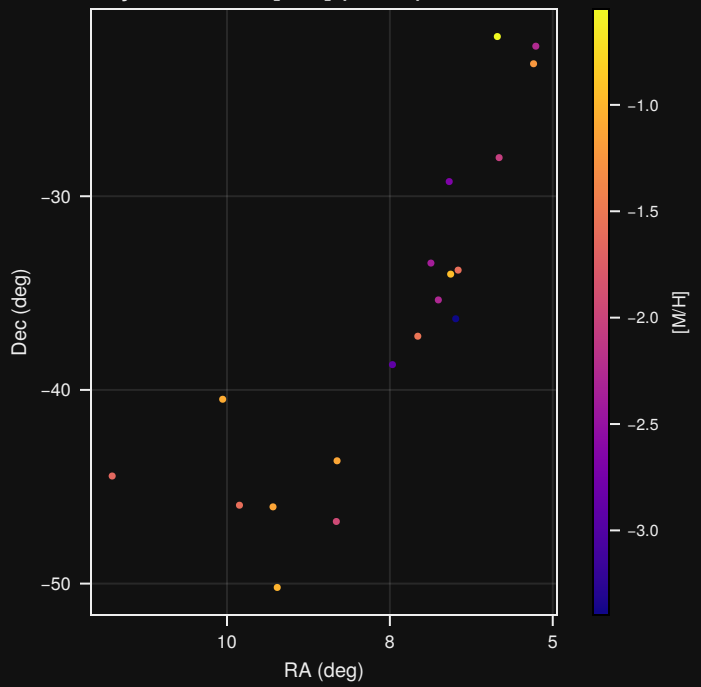
Star types from logg (n = 19)



Kiel diagram (n = 19)



Sky · GSP-Phot [M/H] (n = 19)

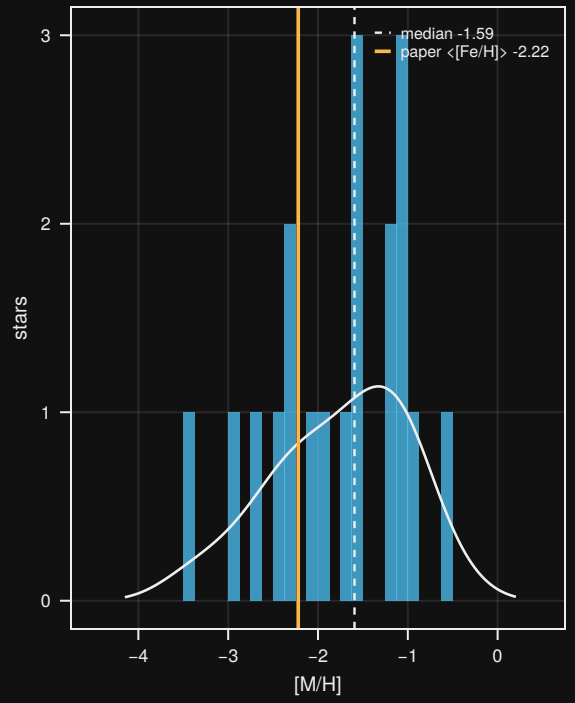


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

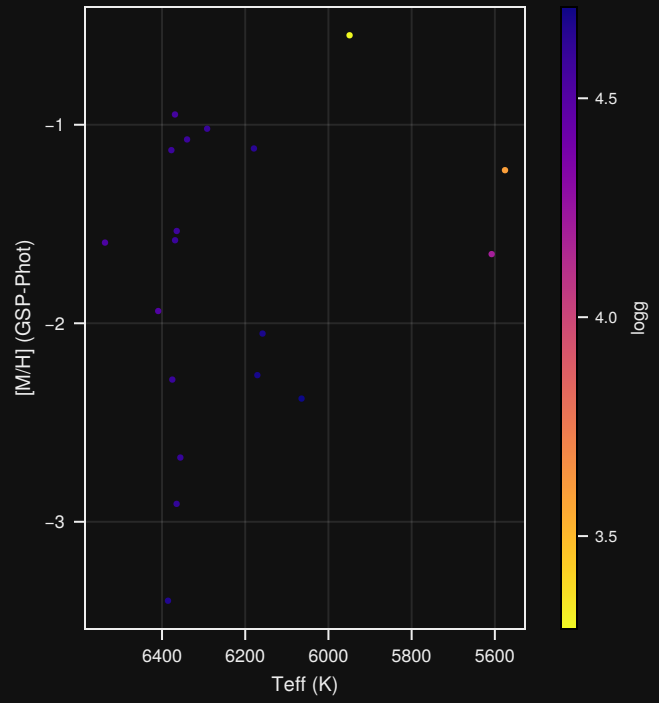
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Metallicity · age · astrometric quality

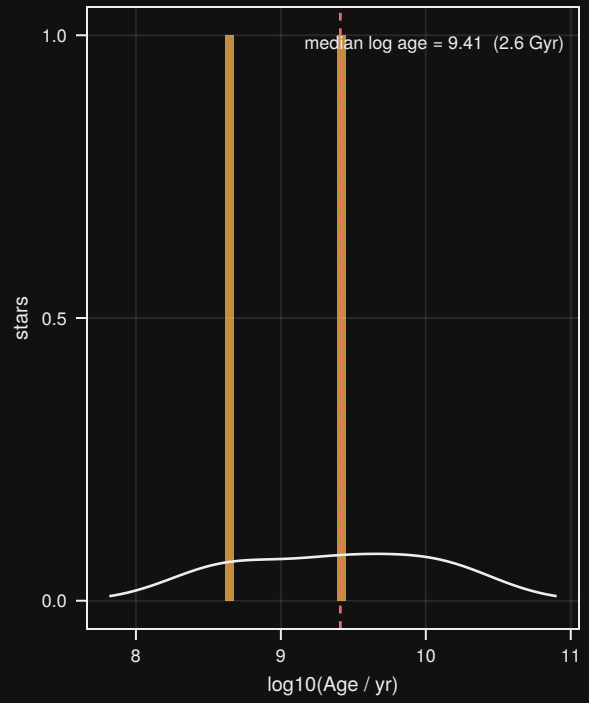
[M/H] GSP-Phot (n = 19)



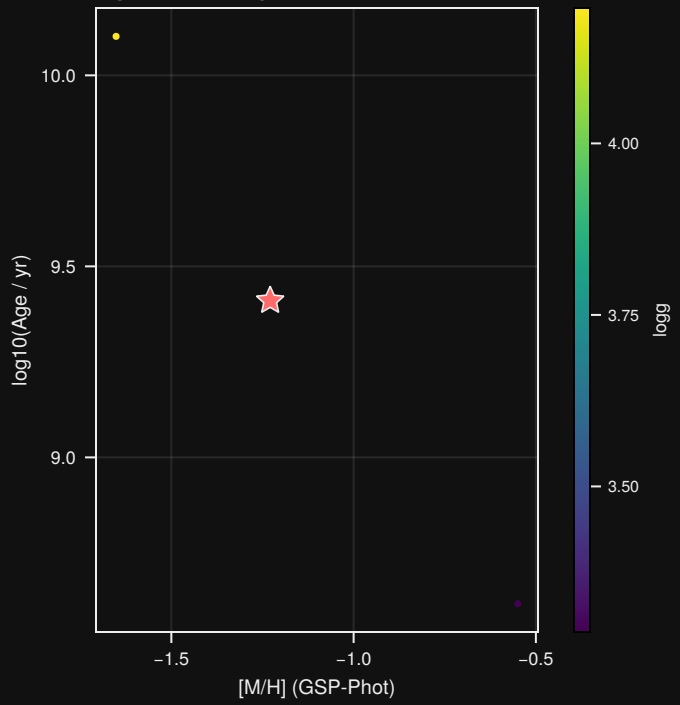
Teff vs [M/H] (n = 19)



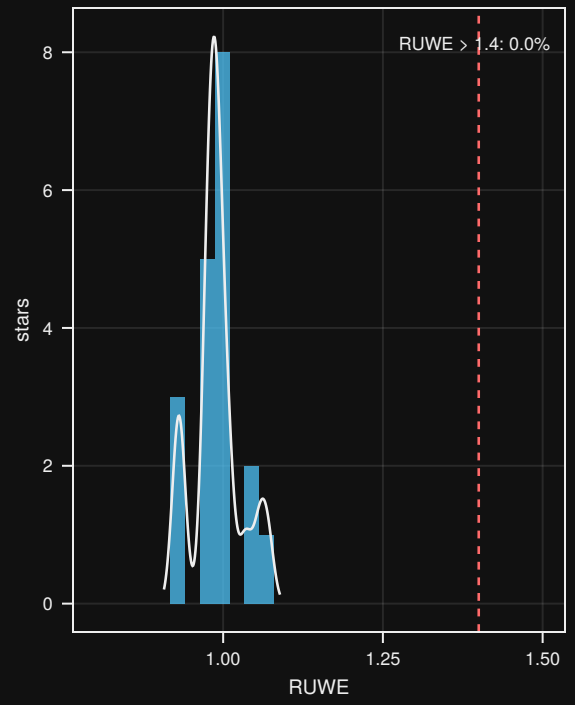
FLAME age (n = 3)



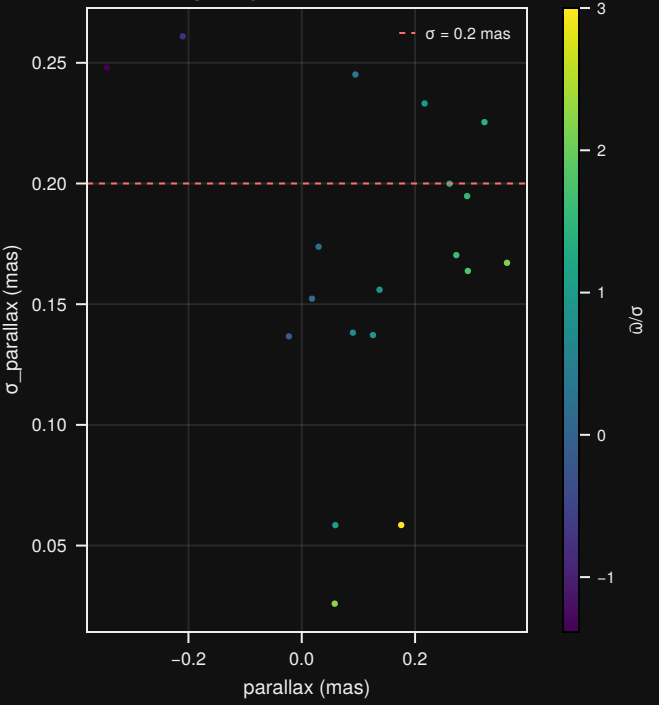
Age–metallicity (n = 3)



RUWE (n = 19)

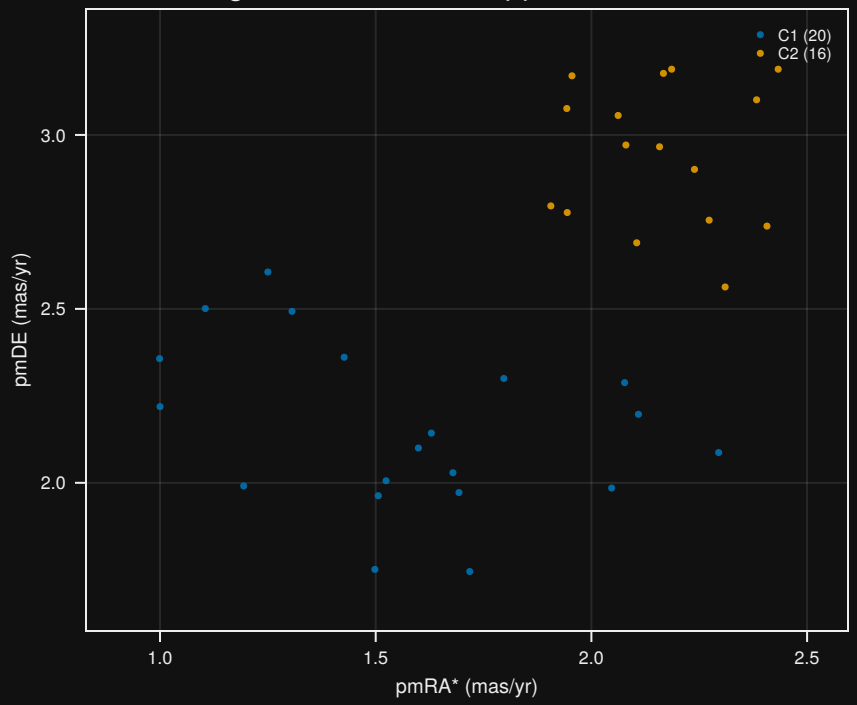


Parallax quality (n = 19)

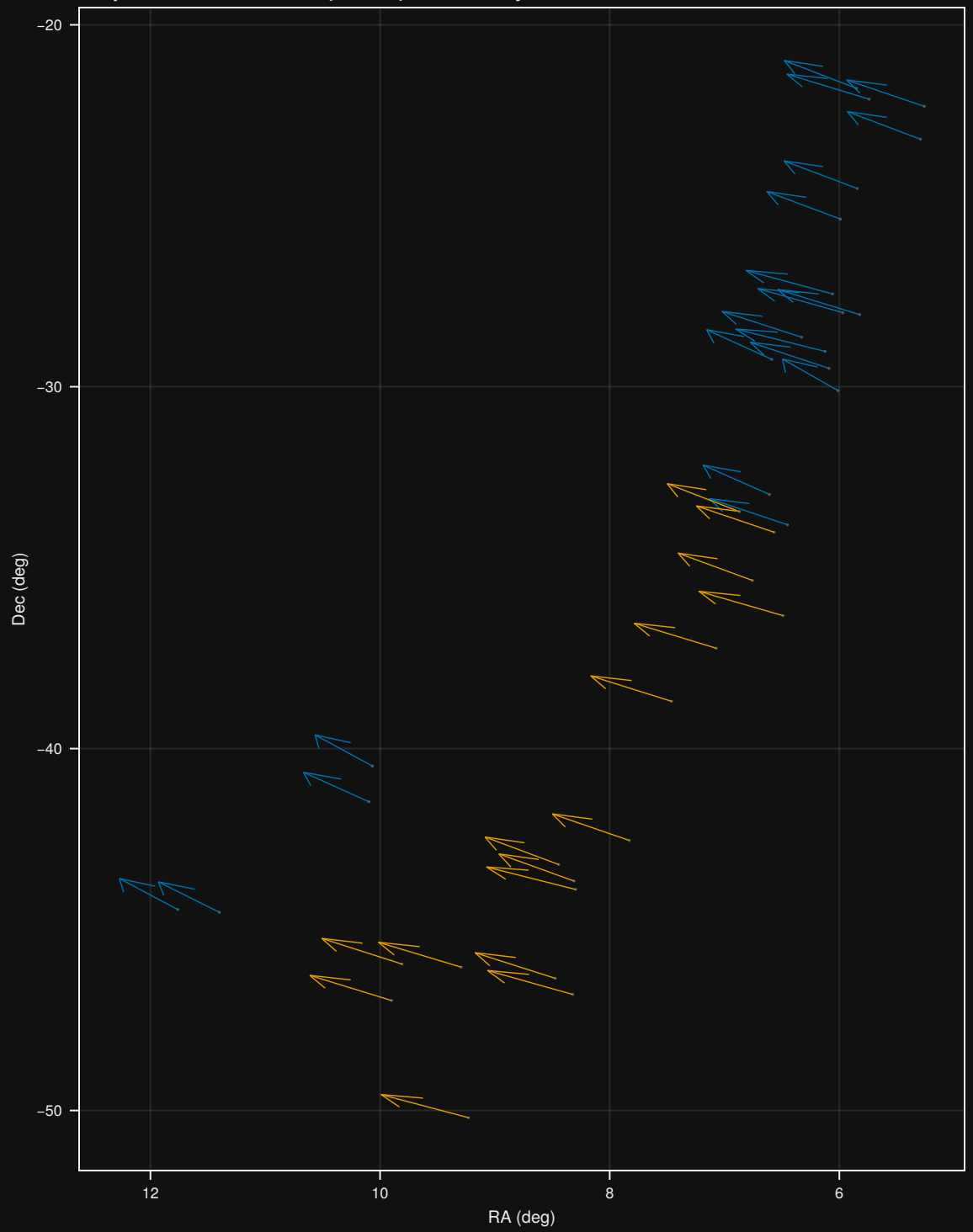


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PM clustering · HDBSCAN · 2 cluster(s) · noise 0%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	0	—	—	—	—	0
C1	20	1.56	2.12	0.36	-1.62	1
C2	16	2.16	2.97	0.19	-1.59	0

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Per-cluster proper-motion vector maps

